

Improve Wasserstein convergence rate for vector martingale central limit theorem

W4414783466_p0 (website page)

Website #32 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

Let $(m_k)_{k \geq 1}$ be a d -dimensional martingale difference sequence with respect to $(\mathcal{F}_k)_{k \geq 0}$, with partial sums $S_n = \sum_{k=1}^n m_k$ and $W_n = S_n/\sqrt{n}$. Write

$$\Sigma_k = \mathbb{E}[m_k m_k^\top \mid \mathcal{F}_{k-1}], \quad Z \sim \mathcal{N}(0, I_d),$$

and let $\Sigma_\infty \succ 0$ be the intended limiting covariance.

The task is to exhibit additional, explicit, checkable hypotheses under which there is a constant $C < \infty$ such that

$$d_W(W_n, \Sigma_\infty^{1/2} Z) \leq \frac{C}{\sqrt{n}} \quad (n \geq 1),$$

thus removing the logarithmic factor that appears in rougher general martingale Wasserstein bounds.

I interpret the question as asking for *sufficient* conditions, not a necessary-and-sufficient characterization.

Alternative readings considered

Two readings seemed possible.

1. Find a theorem for the exact fixed-sequence setting stated in the prompt.
2. Allow a triangular-array reformulation.

I adopt the harder fixed-sequence reading. I also interpret “improved rate $O(1/\sqrt{n})$ ” in the strong non-asymptotic sense: there should exist a constant C independent of n .

Related literature found

- Mourrat (2013) — **PARTIAL**. Scalar martingale CLT in Kantorovich distance; useful background for where logarithmic losses come from, but not a multivariate d_W theorem. (infoscience.epfl.ch)
- Fan–Ma (2020) — **PARTIAL**. Scalar Wasserstein bound for martingales via a Bolthausen/Lindeberg argument; again not vector-valued. (arxiv.org)
- Guo (2024 preprint) — **PARTIAL**. Gives scalar Wasserstein- r rates in terms of Lyapunov coefficients and conditional-variance fluctuation, and explicitly advertises optimal W_1 behavior up to constants; this was useful for choosing the right shape of extra assumptions. (arxiv.org)
- Fang–Shao–Xu (2019) — **TOOL/BACKGROUND**. Multivariate Wasserstein normal approximation by Stein/Bismut methods; relevant because it highlights that the central obstruction is regularity of the test function after smoothing. I do *not* invoke it as a black box. (researchportal.hkust.edu.hk)

- Bonis (2019 preprint / later published) — **BACKGROUND**. Independent multivariate CLT results in Wasserstein distance with improved rates under additional regularity/continuous-component assumptions; this suggested trying a martingale analogue with a built-in Gaussian component. I do *not* invoke it as a black box. (arxiv.org)
- Fan–Su (2025) — **PARTIAL**. Scalar standardized-martingale Wasserstein rates via Stein plus Lindeberg telescoping; relevant methodologically, but not a vector theorem. (sciencedirect.com)

Proof sketch

The logarithmic loss in Wasserstein-1 bounds for martingales is typically a *regularity loss*: one wants to test against a merely Lipschitz function h , but the Lindeberg replacement argument naturally produces second- and third-derivative terms. If one mollifies h externally at a tiny scale, the derivative norms blow up, and the optimization over that scale is where logarithmic losses usually enter.

So I impose a stronger, very concrete Cramér-type smoothing condition: each increment contains a *fresh Gaussian component* of fixed covariance $Q \succ 0$. Then the normalized sum W_n itself contains an n -independent Gaussian convolution $N(0, Q)$. Therefore every 1-Lipschitz test function h is automatically replaced by the smoothed function

$$\Phi_h(x) = \mathbb{E}[h(x + G_Q)], \quad G_Q \sim \mathcal{N}(0, Q),$$

whose second and third derivatives are bounded by constants depending only on Q , not on n .

Once this smoothing is built into the model, one can run a direct multivariate Lindeberg telescoping argument on the latent martingale part (ξ_k) , comparing it to a Gaussian sum with covariance $\Sigma_\infty - Q$. The first-order term cancels by the martingale difference property. The second-order term is exactly the predictable covariance error, and the third-order term is controlled by a uniform third moment. This yields the classical $1/\sqrt{n}$ rate.

The price is that the extra hypothesis is strong: it assumes a fixed nondegenerate Gaussian innovation inside every increment. But it is fully checkable, keeps the vector martingale structure, and gives a clean rigorous mechanism that removes the logarithmic factor.

Main result

Theorem 1. *Let $Q \in \mathbb{R}^{d \times d}$ be deterministic, symmetric, and positive definite, and set*

$$A := \Sigma_\infty - Q.$$

Assume $A \succeq 0$. Suppose there exist:

- *a filtration $(\mathcal{G}_k)_{k \geq 0}$,*
- *a d -dimensional martingale difference sequence $(\xi_k)_{k \geq 1}$ with respect to $(\mathcal{G}_k)$,*
- *i.i.d. standard Gaussian vectors $(\zeta_k)_{k \geq 1}$, independent of $\mathcal{G}_\infty := \sigma(\bigcup_{k \geq 0} \mathcal{G}_k)$,*

such that

$$\mathcal{F}_k = \mathcal{G}_k \vee \sigma(\zeta_1, \dots, \zeta_k), \quad m_k = \xi_k + Q^{1/2} \zeta_k \quad (k \geq 1).$$

Assume further that

$$M_3 := \sup_{k \geq 1} \mathbb{E} \|\xi_k\|^3 < \infty$$

and that there exists $B < \infty$ such that for every $n \geq 1$,

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} \leq \frac{B}{\sqrt{n}}.$$

Let φ_Q be the density of $\mathcal{N}(0, Q)$, and define

$$C_2(Q) := \sum_{i,j=1}^d \int_{\mathbb{R}^d} \|y\| |\partial_{ij} \varphi_Q(y)| dy,$$

$$C_3(Q) := \sum_{i,j,\ell=1}^d \int_{\mathbb{R}^d} \|y\| |\partial_{ij\ell} \varphi_Q(y)| dy,$$

and

$$\nu_3(A) := \mathbb{E} \|A^{1/2} Z\|^3.$$

Then, for every $n \geq 1$,

$$d_W(W_n, \Sigma_\infty^{1/2} Z) \leq \left(\frac{C_2(Q)}{2} B + \frac{C_3(Q)}{6} (M_3 + \nu_3(A)) \right) \frac{1}{\sqrt{n}}.$$

In particular, if $\Sigma_k = \Sigma_\infty$ almost surely for every k , then $B = 0$ and the same assumptions imply the pure rate

$$d_W(W_n, \Sigma_\infty^{1/2} Z) = O(n^{-1/2}).$$

Proof

Lemma 1. For every $h \in \text{Lip}_1$, define

$$\Phi_h(x) := \int_{\mathbb{R}^d} h(x+y) \varphi_Q(y) dy = \mathbb{E}[h(x+G_Q)], \quad G_Q \sim \mathcal{N}(0, Q).$$

Then $\Phi_h \in C^\infty(\mathbb{R}^d)$, and

$$\sup_{x \in \mathbb{R}^d} \|\nabla^2 \Phi_h(x)\|_{HS} \leq C_2(Q),$$

$$\sup_{x \in \mathbb{R}^d} \sup_{\|u\| \leq 1, \|v\| \leq 1, \|w\| \leq 1} |D^3 \Phi_h(x)[u, v, w]| \leq C_3(Q).$$

Proof. Because $Q \succ 0$, the Gaussian density φ_Q is C^∞ , and each second or third derivative of φ_Q is a polynomial times φ_Q . Hence

$$\|y\| |\partial_{ij} \varphi_Q(y)|, \quad \|y\| |\partial_{ij\ell} \varphi_Q(y)|$$

are integrable on $\mathbb{R}^d$, so $C_2(Q)$ and $C_3(Q)$ are finite.

Fix $x \in \mathbb{R}^d$. Since $\int \partial_{ij} \varphi_Q(y) dy = 0$, we may write

$$\partial_{ij} \Phi_h(x) = \int_{\mathbb{R}^d} (h(x+y) - h(x)) \partial_{ij} \varphi_Q(y) dy.$$

Because h is 1-Lipschitz,

$$|h(x+y) - h(x)| \leq \|y\|.$$

Therefore

$$|\partial_{ij}\Phi_h(x)| \leq \int_{\mathbb{R}^d} \|y\| |\partial_{ij}\varphi_Q(y)| dy.$$

Summing over i, j and using that the Hilbert–Schmidt norm is bounded by the sum of the absolute values of the entries gives

$$\|\nabla^2\Phi_h(x)\|_{HS} \leq C_2(Q).$$

The same argument with third derivatives yields

$$\partial_{ij\ell}\Phi_h(x) = \int_{\mathbb{R}^d} (h(x+y) - h(x)) \partial_{ij\ell}\varphi_Q(y) dy,$$

hence

$$|\partial_{ij\ell}\Phi_h(x)| \leq \int_{\mathbb{R}^d} \|y\| |\partial_{ij\ell}\varphi_Q(y)| dy.$$

Now let $u, v, w \in \mathbb{R}^d$ with $\|u\|, \|v\|, \|w\| \leq 1$. Then

$$D^3\Phi_h(x)[u, v, w] = \sum_{i,j,\ell=1}^d u_i v_j w_\ell \partial_{ij\ell}\Phi_h(x),$$

so

$$|D^3\Phi_h(x)[u, v, w]| \leq \sum_{i,j,\ell=1}^d |\partial_{ij\ell}\Phi_h(x)| \leq C_3(Q).$$

This proves the claim. □

Lemma 2. *Let*

$$A_k := \mathbb{E}[\xi_k \xi_k^\top \mid \mathcal{G}_{k-1}].$$

Then

$$\Sigma_k = A_k + Q \quad \text{almost surely for every } k.$$

Hence

$$A_k - A = \Sigma_k - \Sigma_\infty.$$

Proof. Since $m_k = \xi_k + Q^{1/2}\zeta_k$,

$$m_k m_k^\top = \xi_k \xi_k^\top + \xi_k \zeta_k^\top Q^{1/2} + Q^{1/2} \zeta_k \xi_k^\top + Q^{1/2} \zeta_k \zeta_k^\top Q^{1/2}.$$

Condition on $\mathcal{F}_{k-1} = \mathcal{G}_{k-1} \vee \sigma(\zeta_1, \dots, \zeta_{k-1})$. Because ζ_k is independent of $\mathcal{G}_\infty$ and of $\zeta_1, \dots, \zeta_{k-1}$, it is independent of $\mathcal{F}_{k-1}$ and of ξ_k . Also $\mathbb{E}[\zeta_k] = 0$ and $\mathbb{E}[\zeta_k \zeta_k^\top] = I_d$. Therefore the two cross terms have conditional expectation 0, and

$$\mathbb{E}\left[Q^{1/2} \zeta_k \zeta_k^\top Q^{1/2} \mid \mathcal{F}_{k-1}\right] = Q.$$

It remains to identify $\mathbb{E}[\xi_k \xi_k^\top \mid \mathcal{F}_{k-1}]$. Since $\sigma(\zeta_1, \dots, \zeta_{k-1})$ is independent of $\mathcal{G}_\infty$, conditioning on $\mathcal{F}_{k-1}$ gives no more information about ξ_k than conditioning on $\mathcal{G}_{k-1}$. Hence

$$\mathbb{E}[\xi_k \xi_k^\top \mid \mathcal{F}_{k-1}] = \mathbb{E}[\xi_k \xi_k^\top \mid \mathcal{G}_{k-1}] = A_k.$$

So $\Sigma_k = A_k + Q$. Since $\Sigma_\infty = A + Q$, subtracting yields

$$A_k - A = \Sigma_k - \Sigma_\infty. \quad \square$$

Lemma 3. Let $Y_1, \dots, Y_n$ be i.i.d. $\mathcal{N}(0, A)$, independent of everything else, and put

$$U_n := \frac{1}{\sqrt{n}} \sum_{k=1}^n \xi_k, \quad V_n := \frac{1}{\sqrt{n}} \sum_{k=1}^n Y_k.$$

Then for every $h \in \text{Lip}_1$,

$$|\mathbb{E}[\Phi_h(U_n)] - \mathbb{E}[\Phi_h(V_n)]| \leq \frac{C_2(Q)}{2n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} + \frac{C_3(Q)}{6n^{3/2}} \sum_{k=1}^n (\mathbb{E} \|\xi_k\|^3 + \mathbb{E} \|Y_k\|^3).$$

Proof. Define the telescoping sequence

$$T_k := \frac{1}{\sqrt{n}} \left(\sum_{i=1}^k \xi_i + \sum_{i=k+1}^n Y_i \right), \quad 0 \leq k \leq n.$$

Then $T_0 = V_n$ and $T_n = U_n$, so

$$\mathbb{E}[\Phi_h(U_n)] - \mathbb{E}[\Phi_h(V_n)] = \sum_{k=1}^n \mathbb{E}[\Phi_h(T_k) - \Phi_h(T_{k-1})].$$

For each k , write

$$R_k := \frac{1}{\sqrt{n}} \left(\sum_{i=1}^{k-1} \xi_i + \sum_{i=k+1}^n Y_i \right),$$

so that

$$T_k = R_k + \frac{\xi_k}{\sqrt{n}}, \quad T_{k-1} = R_k + \frac{Y_k}{\sqrt{n}}.$$

Let

$$\mathcal{H}_{k-1} := \mathcal{G}_{k-1} \vee \sigma(Y_{k+1}, \dots, Y_n).$$

Then R_k is $\mathcal{H}_{k-1}$ -measurable. Also ξ_k and Y_k are independent of $Y_{k+1}, \dots, Y_n$, and

$$\mathbb{E}[\xi_k \mid \mathcal{H}_{k-1}] = \mathbb{E}[\xi_k \mid \mathcal{G}_{k-1}] = 0,$$

$$\mathbb{E}[Y_k \mid \mathcal{H}_{k-1}] = 0,$$

$$\mathbb{E}[\xi_k \xi_k^\top \mid \mathcal{H}_{k-1}] = A_k, \quad \mathbb{E}[Y_k Y_k^\top \mid \mathcal{H}_{k-1}] = A.$$

Apply Taylor's theorem at the point R_k to the function Φ_h . For any vector $x \in \mathbb{R}^d$,

$$\Phi_h(R_k + x) = \Phi_h(R_k) + \nabla \Phi_h(R_k) \cdot x + \frac{1}{2} x^\top \nabla^2 \Phi_h(R_k) x + r_k(x),$$

where

$$|r_k(x)| \leq \frac{C_3(Q)}{6} \|x\|^3$$

by the previous lemma.

Insert $x = \xi_k/\sqrt{n}$ and $x = Y_k/\sqrt{n}$, subtract, and condition on $\mathcal{H}_{k-1}$. The zeroth-order terms cancel. The first-order terms cancel because both conditional means are zero. Hence

$$\mathbb{E}[\Phi_h(T_k) - \Phi_h(T_{k-1}) \mid \mathcal{H}_{k-1}] = \frac{1}{2n} \text{tr}((A_k - A) \nabla^2 \Phi_h(R_k)) + \rho_k,$$

with

$$|\rho_k| \leq \frac{C_3(Q)}{6n^{3/2}} (\mathbb{E}[\|\xi_k\|^3 \mid \mathcal{H}_{k-1}] + \mathbb{E}[\|Y_k\|^3 \mid \mathcal{H}_{k-1}]).$$

Take expectations. By the Hilbert–Schmidt Cauchy–Schwarz inequality and the bound on $\nabla^2 \Phi_h$,

$$\left| \mathbb{E} \left[\frac{1}{2n} \operatorname{tr}((A_k - A) \nabla^2 \Phi_h(R_k)) \right] \right| \leq \frac{C_2(Q)}{2n} \mathbb{E} \|A_k - A\|_{HS}.$$

By Lemma 2,

$$\mathbb{E} \|A_k - A\|_{HS} = \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS}.$$

Also $\mathbb{E} \|Y_k\|^3 = \nu_3(A)$. Therefore

$$|\mathbb{E}[\Phi_h(T_k) - \Phi_h(T_{k-1})]| \leq \frac{C_2(Q)}{2n} \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} + \frac{C_3(Q)}{6n^{3/2}} (\mathbb{E} \|\xi_k\|^3 + \nu_3(A)).$$

Summing over $k = 1, \dots, n$ proves the lemma. $\square$

Proof of the theorem. Let

$$G_n := \frac{1}{\sqrt{n}} \sum_{k=1}^n Q^{1/2} \zeta_k.$$

Because the ζ_k are i.i.d. $\mathcal{N}(0, I_d)$, we have

$$G_n \sim \mathcal{N}(0, Q).$$

Since the whole Gaussian sequence (ζ_k) is independent of $\mathcal{G}_\infty$, G_n is independent of U_n . Thus

$$W_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n m_k = \frac{1}{\sqrt{n}} \sum_{k=1}^n \xi_k + \frac{1}{\sqrt{n}} \sum_{k=1}^n Q^{1/2} \zeta_k = U_n + G_n.$$

Now fix $h \in \operatorname{Lip}_1$. By definition of Φ_h ,

$$\mathbb{E}[h(W_n)] = \mathbb{E}[h(U_n + G_n)] = \mathbb{E}[\Phi_h(U_n)].$$

On the other hand, if $V_n \sim \mathcal{N}(0, A)$ and $G_n \sim \mathcal{N}(0, Q)$ are independent, then $V_n + G_n \sim \mathcal{N}(0, A + Q) = \mathcal{N}(0, \Sigma_\infty)$, so

$$\mathbb{E}[h(\Sigma_\infty^{1/2} Z)] = \mathbb{E}[\Phi_h(V_n)].$$

Therefore

$$|\mathbb{E}[h(W_n)] - \mathbb{E}[h(\Sigma_\infty^{1/2} Z)]| = |\mathbb{E}[\Phi_h(U_n)] - \mathbb{E}[\Phi_h(V_n)]|.$$

Applying Lemma 3 gives

$$|\mathbb{E}[h(W_n)] - \mathbb{E}[h(\Sigma_\infty^{1/2} Z)]| \leq \frac{C_2(Q)}{2n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} + \frac{C_3(Q)}{6n^{3/2}} \sum_{k=1}^n (\mathbb{E} \|\xi_k\|^3 + \nu_3(A)).$$

By assumption,

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} \leq \frac{B}{\sqrt{n}},$$

and

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \|\xi_k\|^3 \leq M_3.$$

Hence

$$|\mathbb{E}[h(W_n)] - \mathbb{E}[h(\Sigma_\infty^{1/2} Z)]| \leq \left(\frac{C_2(Q)}{2} B + \frac{C_3(Q)}{6} (M_3 + \nu_3(A)) \right) \frac{1}{\sqrt{n}}.$$

Taking the supremum over all $h \in \text{Lip}_1$ yields

$$d_W(W_n, \Sigma_\infty^{1/2} Z) \leq \left(\frac{C_2(Q)}{2} B + \frac{C_3(Q)}{6} (M_3 + \nu_3(A)) \right) \frac{1}{\sqrt{n}}.$$

This proves the theorem. □

Gap to the full statement

The full problem asks for useful additional conditions removing the logarithm in a vector martingale Wasserstein CLT. I proved this under a *strong* but clean smoothing hypothesis:

- every increment contains a fresh nondegenerate Gaussian component of fixed covariance $Q \succ 0$;
- the latent martingale part has uniformly bounded third moments;
- the average predictable covariance error satisfies

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} = O(n^{-1/2}).$$

What remains unproved here is the much broader claim that one can remove the logarithm under *only* natural moment and covariance assumptions, without a built-in smoothing component. My proof identifies the precise obstruction: without such smoothing, the needed second and third derivative bounds for Lipschitz test functions are no longer uniform in n , and one must solve a genuinely rough multivariate Stein-regularity problem.

Self red-team

Counterexample attempts.

1. I tried to drop the Gaussian component ($Q = 0$). Then $C_2(Q)$ and $C_3(Q)$ become meaningless: the convolution regularization disappears. This is not a bug of the proof; it is exactly where the logarithmic loss is expected to re-enter.
2. I tried to weaken the covariance assumption from

$$\frac{1}{n} \sum_{k=1}^n \mathbb{E} \|\Sigma_k - \Sigma_\infty\|_{HS} = O(n^{-1/2})$$

to merely

$$\mathbb{E} \left\| \frac{1}{n} \sum_{k=1}^n (\Sigma_k - \Sigma_\infty) \right\|_{HS} = O(n^{-1/2}).$$

My telescoping proof does *not* support that weaker form, because the Hessian is evaluated at different random points R_k . I therefore did not claim it.

3. I checked the degenerate case $A = 0$, i.e. $\Sigma_\infty = Q$. Then the latent Gaussian comparator V_n is identically 0, and the proof still works. In the fully Gaussian case $\xi_k \equiv 0$, the bound collapses to 0, as it should.

Three likely referee objections and responses.

1. *Objection: do future Gaussian comparator terms accidentally depend on the current martingale increment?*

Response: no. I deliberately compare ξ_k to an i.i.d. Gaussian $Y_k \sim N(0, A)$ with *deterministic* covariance $A = \Sigma_\infty - Q$. A tempting but flawed variant would use $Y_k \sim N(0, A_k)$; that fails because future A_i depend on past martingale increments. I do not use that flawed construction.

2. *Objection: is differentiation under the integral sign legitimate when h is only Lipschitz and unbounded?*

Response: yes, because I never differentiate h . I differentiate the Gaussian density and use

$$\partial^\alpha \Phi_h(x) = \int (h(x+y) - h(x)) \partial^\alpha \varphi_Q(y) dy,$$

where $\|y\| \partial^\alpha \varphi_Q(y) \in L^1$ for $|\alpha| = 2, 3$. This is exactly why the Gaussian smoothing assumption is so convenient.

3. *Objection: did you secretly use an external Stein theorem?*

Response: no. The proof uses only Taylor expansion, Gaussian convolution identities, and elementary conditioning. The cited literature was used only to select a promising strategy.

Boundary checks.

- $d = 1$: the theorem reduces to a scalar smoothed martingale Berry–Esseen statement.
- Exact bracket case $\Sigma_k = \Sigma_\infty$: the second-order term vanishes and the rate is purely $O(n^{-1/2})$.
- Fully Gaussian increments: the theorem gives error 0.

Ideas tried

1. **Direct multivariate Stein equation for Lipschitz tests without extra smoothing.** Idea: solve the Gaussian Stein equation for $h \in \text{Lip}_1$ and run a martingale exchange/Taylor argument directly. Obstruction: for rough h , third-derivative control is precisely where logarithmic losses enter. Next step: one would need a genuinely sharp multivariate Bismut/Malliavin estimate adapted to martingale increments.
2. **Gaussian comparator with random covariance A_k .** Idea: compare ξ_k to $Y_k \sim \mathcal{N}(0, A_k)$, so the second-order term cancels exactly. Obstruction: future Y_i then depend on ξ_k through A_i , so the clean conditioning step breaks. Next step: replace the naive telescoping by a martingale Stein interpolation that handles random predictable covariances globally.
3. **Reduction to one-dimensional projections.** Idea: apply scalar martingale Wasserstein bounds to $\langle u, W_n \rangle$ for every $u \in S^{d-1}$. Obstruction: I do not know a sharp, clean, fixed-dimension inequality that upgrades uniform scalar W_1 control over all directions to full Euclidean W_1 control for general vector laws. Next step: find or prove a suitable Cramér–Wold-to- W_1 lifting result.

4. **Incrementwise coupling to Gaussian increments.** Idea: couple each m_k directly to a Gaussian with covariance Σ_k . Obstruction: summing per-step W_1 errors gives an $O(\sqrt{n})$ upper bound after normalization, which ignores CLT cancellation entirely. Next step: combine couplings with a second-order cancellation argument, not a raw triangle inequality.

Invoked results

No external theorem is invoked as a black-box proof step.

The proof uses only:

- Taylor’s theorem with third-order remainder;
- the fact that convolution with a Gaussian density regularizes Lipschitz functions;
- the elementary fact that sums of independent centered Gaussian vectors are Gaussian with covariance equal to the sum of the individual covariances.

All of these are derived or used directly in-line. The papers listed in the literature section are **BACKGROUND** or **PARTIAL** only and are not imported into the proof.

Self-confidence

3 — I checked the telescoping, conditioning, and smoothing estimates carefully, and I could not break the proof; my main residual caution is that the filtration/product-structure assumption is strong and must be read exactly as stated.

Bibliography

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\begin{verbatim}
@article{Mourrat2013,
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  year = {2013},
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@article{FanMa2020,
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@article{Guo2024,
  author = {Guo, Xiaoqin},
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@article{FangShaoXu2019,
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    title = {Multivariate approximations in Wasserstein distance by Stein's method and Bismut's
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    year = {2019},
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@article{Bonis2019,
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@article{FanSu2025,
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